

AJAY VENKATESH

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Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

• CampusMesh

Jan 2026 – Present

AI Software Engineer

New York, US

- Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
- Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda**, **API Gateway**, **S3**, and **CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

• Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
- Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate**, **SQS**, **SNS**, decoupling distributed message processing and reducing response latency by **50%**.
- Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.

• New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

AI Software Engineer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
- Built end-to-end **computer vision pipelines (YOLOv8, EfficientNet)** for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
- Developed a full-stack insurance analytics product (**React**, **Node.js**, **REST APIs**) integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.

• PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Designed a distributed backend on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting **ML inference pipelines** for healthcare analytics.
- Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and **Azure Form Recognizer** to automate document data extraction, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** on **100K+** records, reducing execution from **5 hours to 1 hour**.
- Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

Projects

• InviLite (AI-Driven Data Engineering & Analytics Platform) | AWS, SageMaker, Bedrock, FAISS, Python

- Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.

• RoBERTa Fine-Tuning with Custom LoRA | Python, PyTorch, Hugging Face, AGNews

- Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
- Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.

• High-Efficiency ResNet Image Classifier | Python, PyTorch, Torchvision, Matplotlib

- Designed and implemented a specialized **ResNet** architecture optimized for the **CIFAR-10** dataset, adhering to a strict **5M parameter** limit for edge-ready deployment.
- Enhanced model convergence by integrating the **Lookahead optimizer** and advanced data augmentation, with real-time visualization via **TensorBoard**.

Skills

- **Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL, Rust
- **AI/ML:** PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Computer Vision (YOLOv8, EfficientNet), Time-Series Modeling, MLflow, SVM, Random Forest, Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Cursor, GitHub Copilot, Model Context Protocol (MCP), Robotics, CUDA

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals